

Tying the Data Knot

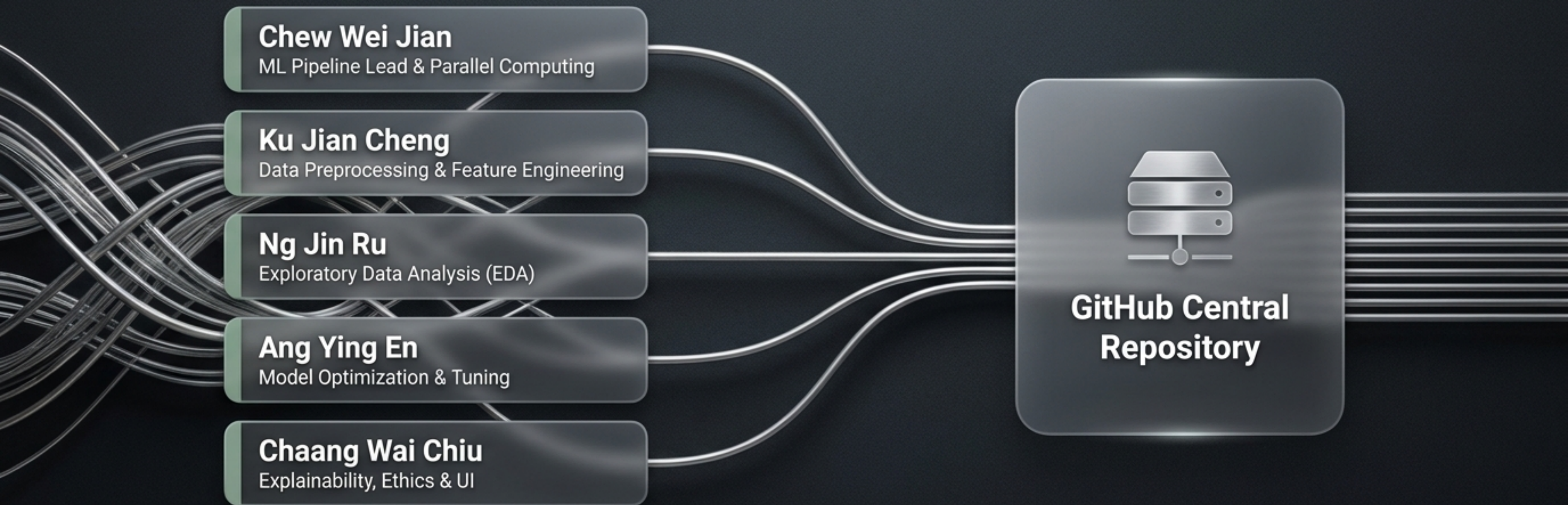
A State-of-the-Art Machine
Learning Pipeline for Digital
Romance (V8 Patched Edition)

OCC6 Group 3

WIA1006 Machine Learning

Faculty of Computer Science and
Information Technology

Team Organization & Computing Architecture



Execution Architecture: Modular Jupyter Notebook Branches | Parallel Execution | Strict Peer-Review Protocol

The Hypothesis & Stakes



The Problem

Superficial
Matchmaking

Confounding
Sociological Factors

High Churn &
Ghosting Rates



The Objective

Transitioning from
Predictive to Prescriptive AI

Build a Leakage-Free
Causal Audit Sandbox

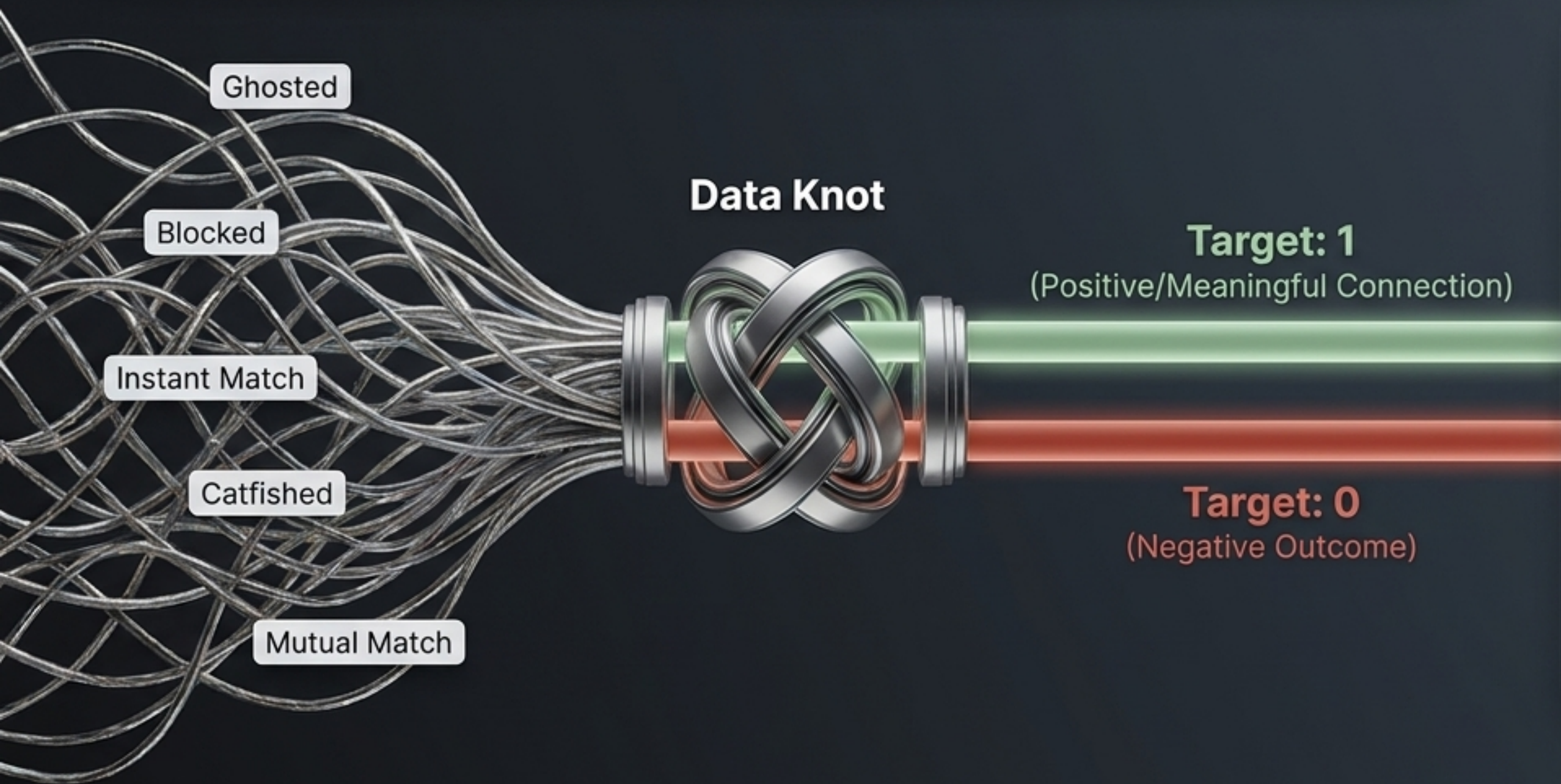
Determine Why Matches
Happen, Not Just If

The Dataset & Target Variable

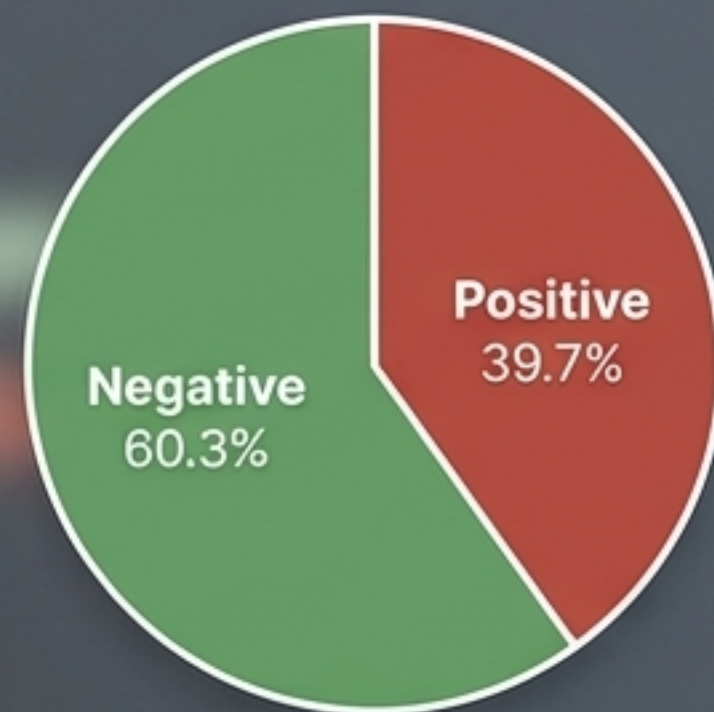
50,000 Samples

25 Features

Zero Missing Values

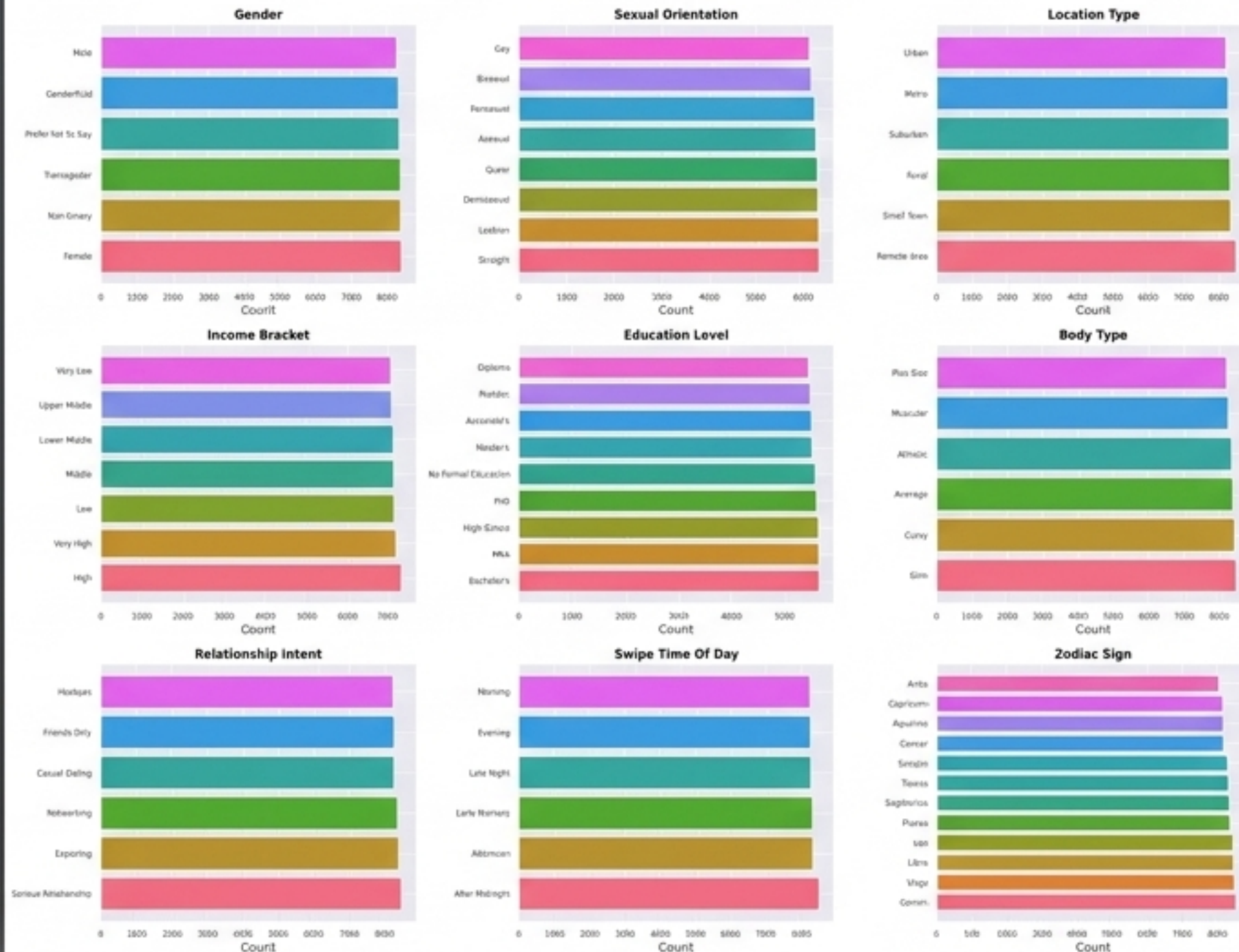


Binary Target Distribution
(Positive vs Negative Outcome)



Exploratory Data Analysis: Categorical Uniformity

Categorical Feature Distributions



Programmatic Uniformity

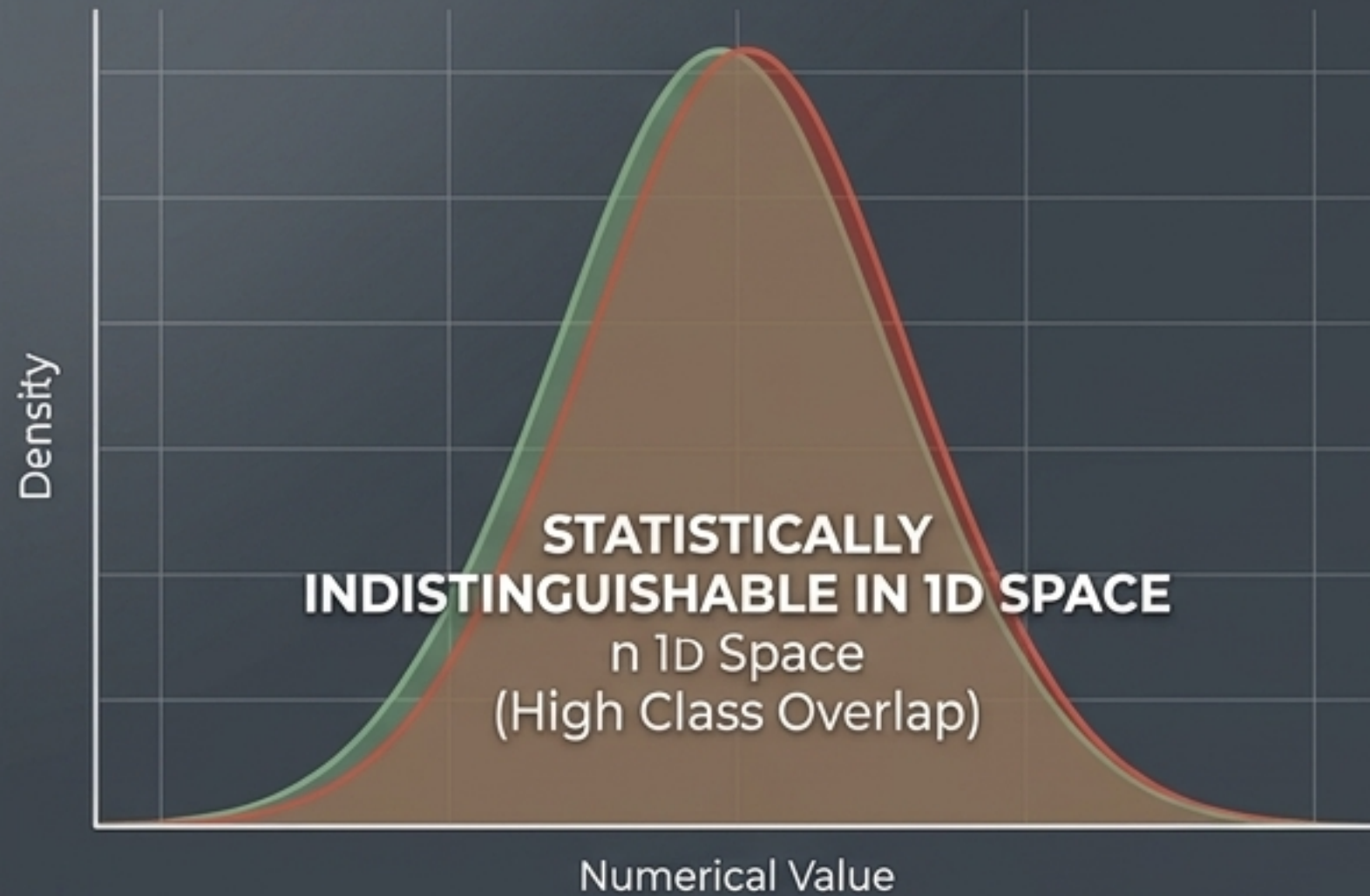
Uniform distributions across sensitive demographic variables.

Prevents class dominance in the algorithm's loss function.

Establishes a pristine, mathematically sound baseline for Demographic Parity Fairness Audits.

Exploratory Data Analysis: Numerical Density & Outliers

Numerical Feature Density Overlap



Outlier Detection — Boxplots



METHODOLOGICAL CONSEQUENCE: UPGRADE TO MEDIAN-BASED ROBUSTSCALER
(Deferred Post-Split to Prevent Leakage)